

MAINE-SOROA, Niger

WMO: 610960

Lat: 13.233N

Lon: 11.983E

Elev: 1109

StdP: 14.12

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.1	57.4	17.2	13.7	76.7	17.6	13.9	77.6	18.0	72.7	16.3	72.9	3.3	40	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	23.8	108.5	69.9	107.0	69.7	105.3	69.8	79.8	92.7	78.9	91.9	78.2	91.0	4.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.8	145.4	84.8	75.9	141.1	83.9	75.2	137.6	83.1	44.2	92.9	43.3	91.9	42.5	91.2	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.2	13.3	11.0	DB	50.5	111.7	1.5	1.2	49.4	112.6	48.5	113.3	47.7	114.0	46.6	114.9
			WB	39.0	83.4	1.3	2.8	38.0	85.4	37.3	87.1	36.5	88.7	35.6	90.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.1	72.2	78.0	85.8	92.2	93.7	91.2	86.0	82.7	85.6	87.3	80.8	73.5
	DBStd	7.77	5.07	5.89	4.62	3.66	3.15	3.82	3.94	3.22	3.17	2.52	3.99	4.09
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	2	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12446	689	785	1109	1265	1355	1235	1116	1015	1069	1156	924	728
	CDD65	6974	226	365	644	815	890	785	651	550	619	691	474	264
	CDH74	95037	2555	4470	9167	12921	14206	11788	8333	5989	7558	9389	5817	2844
	CDH80	56936	1124	2424	5759	8855	9826	7613	4445	2654	4036	5661	3300	1239
Wind	WSAvg	4.6	5.1	5.2	5.2	4.5	4.7	5.5	5.2	3.9	3.4	3.6	4.6	4.6
Precipitation	PrecAvg	13.8	0.0	0.0	0.0	0.1	0.3	1.2	4.3	5.9	2.1	0.4	0.0	0.0
	PrecMax	25.6	0.0	0.0	0.2	1.6	1.7	3.6	8.9	17.2	6.1	3.4	0.2	0.2
	PrecMin	6.5	0.0	0.0	0.0	0.0	0.0	0.0	1.2	0.0	0.1	0.0	0.0	0.0
	PrecStd	4.5	0.0	0.0	0.0	0.3	0.4	0.9	2.0	3.1	1.3	0.7	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.0	104.1	107.1	110.5	110.6	107.8	103.2	98.0	103.4	104.3	101.9	97.9
		MCWB	62.0	64.2	66.3	68.9	70.7	73.0	75.0	77.3	74.1	70.0	66.0	63.4
	2%	DB	94.1	100.6	105.1	108.7	108.8	105.7	100.2	95.1	100.4	102.7	99.5	93.5
		MCWB	60.2	63.5	65.2	68.6	70.8	73.4	75.7	77.7	74.8	69.6	64.9	61.4
	5%	DB	90.5	97.1	102.8	107.1	107.2	103.7	97.8	92.9	98.1	101.3	97.3	90.4
		MCWB	58.6	61.9	64.2	68.4	70.7	73.9	75.9	77.7	75.9	69.9	63.7	59.4
	10%	DB	86.4	93.2	100.1	105.1	105.4	101.5	95.2	90.7	95.6	99.3	94.8	87.3
		MCWB	56.7	60.1	62.8	68.2	71.2	74.2	75.9	77.5	76.5	69.8	62.7	58.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.5	67.6	72.4	77.1	79.4	79.9	80.1	80.9	80.8	78.7	70.6	64.7
		MCDB	95.2	99.7	97.2	100.8	100.2	98.2	92.9	91.2	93.7	94.7	96.7	95.8
	2%	WB	61.0	64.5	68.0	75.3	77.4	78.1	78.8	79.7	79.4	77.1	66.5	62.2
		MCDB	92.2	97.2	100.2	99.6	98.7	95.8	92.2	90.3	92.2	92.8	96.3	91.4
	5%	WB	59.0	62.5	65.5	73.7	76.3	77.2	77.9	78.8	78.6	76.0	64.7	60.4
		MCDB	89.1	94.6	99.3	97.3	97.4	94.8	91.1	89.4	91.1	91.2	95.2	88.5
	10%	WB	57.1	60.5	63.7	72.0	75.3	76.4	77.1	78.0	77.8	75.0	63.1	58.6
		MCDB	85.5	92.4	98.4	96.5	95.9	93.9	90.1	87.9	89.9	89.6	93.2	85.5
Mean Daily Temperature Range	5% DB	MDBR	28.1	28.5	28.0	26.1	23.8	20.4	17.7	16.0	20.0	25.0	28.0	27.9
		MCDBR	32.3	31.5	30.4	27.5	26.0	23.0	20.7	18.9	23.5	27.2	29.4	30.9
		MCWBR	14.4	13.5	12.6	11.5	11.0	7.5	5.7	6.0	6.8	9.7	12.2	13.7
	5% WB	MCDBR	31.8	30.7	29.0	23.3	22.6	20.0	18.1	17.1	20.2	23.0	28.9	29.7
		MCWBR	14.6	13.8	12.8	10.3	9.1	6.9	6.0	6.6	7.2	8.1	12.9	13.6
Clear-Sky Solar Irradiance	taub	0.541	0.602	0.649	0.749	0.726	0.704	0.667	0.619	0.624	0.664	0.583	0.580	
	taud	1.725	1.593	1.514	1.390	1.444	1.534	1.654	1.756	1.727	1.583	1.639	1.648	
	Ebn at Noon	240	232	225	203	204	206	215	228	228	214	226	224	
	Edh at Noon	72	86	95	107	100	90	80	73	75	86	78	76	
All-Sky Solar Radiation	RadAvg	1814	1950	2085	2134	2118	2007	1878	1819	1951	1933	1854	1756	
	RadStd	48	43	91	72	83	77	56	79	43	59	50	48	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page